

Automata and Computability (Quiz 4)

Course Code: CSE331

Total Marks: 20

Section 17

ID: _____ Name: _____ Class Roll: _____

Questions

Q1. (5+5 Marks) Let $\Sigma = \{0, 1\}$. Write CFG for the following languages.

- $L = \{x, y, w \in \Sigma^* : w = x0y, \text{ where } |x| = |y|\}$
- $L = \{w \in \Sigma^* : w = 0^n 1^m, \text{ where } m = 2n + 1, n \geq 0\}$

Q2. (5+5 Marks) Let $\Sigma = \{0, 1\}$. Draw PDA for the following languages.

- $L = \{w \in \Sigma^* : w \text{ does not have equal 0's and 1's}\}$
- $L = \{w \in \Sigma^* : \text{where } w \text{ is not a palindrome}\}$

Q3. (5 Marks) Short Questions (True / False)

- The number of strings generated by a CFG is always infinite.
- A PDA can have multiple accepting states.
- The number of elements pushed in the stack is dependent on the length of the input.
- When a string is accepted, a stack must be empty.
- PDA's are closed under concatenation.